21-242: Section B, a running problem list

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I'm collating every problem I give, discuss, propose, suggest, or think of in this massive document. I can't promise solutions to all of these, but the more problems, the better. Go nuts! These problems are listed in topic order. It is possible that some problems, even early ones, are listed that you do not yet have the knowledge to prove. Usually it will be somewhat clear if this is the case. That being said, the problem with a large running list is that difficulty isn't disambiguated. When in doubt, check the notes.

1 Preliminary notions: sets and logic

1. Let A, B be sets. Prove that $(A \cup B) \setminus B = A$ if and only if $A \cap B = \varnothing$.

2. Use extensionality / double containment to show that:

   (a) every set is equal to one with duplicate elements removed,

   (b) every set is equal to one with its elements listed in a different order,

   (c) there is exactly one set with no elements.

3. Let X be a set, and for φ a logical formula on X define the set

$$S_\varphi = \{x \in X \mid \varphi(x)\}.$$

   If p, q are logical formulae on X, describe the following in terms of S_p, S_q, and set operations (with proof).

   (a) $S_{\neg p}$

   (b) $S_{p \wedge q}$

   (c) $S_{p \to q}$

   (d) $S_\perp$, where $\perp$ is the logical formula which is always false.

4. *Russell's paradox.* The Axiom of Foundation states that no set contains itself, or rather that there is no infinite decreasing sequence of sets $A_1 \ni A_2 \ni A_3 \ni \cdots$. Let $\mathcal{U}$ denote the class of all sets. *Without* appealing to the Axiom of Foundation, prove that $\mathcal{U}$ cannot be a set.

5. A bunch of sets X and maps f are listed below. Determine which of the following are well-defined operations on their corresponding sets, and find their arity if so.

   (a) $X = \mathbb{Q}$, $f(x, y) = x + y$

   (b) $X = \mathbb{N}$, $f(x, y) = x + y$

   (c) $X = \mathbb{N}$, $f(x, y) = x - y$

   (d) $X = \{0, 1, 2, \ldots, n\}$, $f(x) = n - x$

   (e) $X = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ has an inverse}$, $F(f, g) = f \circ g$

   (f) $X = \mathbb{Q}$, $f(x, y, z, w) = xy - zw$.

6. *The powerset with symmetric difference.* Given a set X and $A, B \in \mathcal{P}(X)$, the *symmetric difference* $A \triangle B$ is defined such that

$$x \in A \triangle B \iff x \in A \text{ or } x \in B, \text{ but not both.}$$

   (a) Show that $\triangle$ is a binary operation on $\mathcal{P}(X)$, and convince yourself that it is commutative.

   (b) Prove, by double containment or a truth table, that $\triangle$ is associative.

   (c) Find all identity elements for $\triangle$.

   (d) Does every set have an inverse under $\triangle$? Prove it, or find a counterexample.

7. A *permutation* of $[n] := \{1, 2, \ldots, n\}$ is a bijective map $\sigma : [n] \to [n]$; denote the collection of all permutations of $[n]$ by S_n. Assume $n \geq 3$. Prove that the map $[\cdot, \cdot] : S_n \times S_n \to S_n$ given by

$$[\sigma, \tau] = \sigma \circ \tau \circ \sigma^{-1} \circ \tau^{-1}$$

   is neither commutative nor associative.

2 Fields and vector spaces

1. *An exercise in number theory.* The purpose of this exercise is to prove that the integers modulo p, $\mathbb{Z}/p\mathbb{Z}$, form a field when p is prime.

   (a) Let $m, n \in \mathbb{Z}$. Prove that there are $a, b \in \mathbb{Z}$ with $am + bn = \gcd(m, n)$. (*Bézout*)

   (b) If p is prime, show that for every $a \in \mathbb{Z}$ not divisible by p there is $b \in \mathbb{Z}$ with $ab = kp + 1$ for some integer k.

   (c) Deduce that the integers modulo p form a field under addition and multiplication modulo p.

   (d) Prove that $\mathbb{Z}/n\mathbb{Z}$ is *not* a field if n is composite.

2. Let $E \subseteq F$ be fields. Show that F forms a vector space over E. (The dimension of this vector space is called the *degree* of F over E, often written $[F : E]$. This is not testable.)

3. Let $(V, +, \cdot)$ a vector space over some field K, W a set, $f : V \to W$ bijective. Define $\oplus : W \times W \to W$ and $\odot : K \times W \to W$ via
$$w \oplus \bar{w} = f(f^{-1}(w) + f^{-1}(\bar{w})),$$
$$\alpha \odot w = f(\alpha f^{-1}(w)).$$
   Prove that $(W, \oplus, \odot)$ is a vector space over K.

4. (a) Let $(V, +)$ be a structure where $+ : V \times V \to V$ is commutative, associative, has identity 0, and satisfies $v + v = 0$ for every $v \in V$. Prove that there is exactly one scalar multiplication $\odot : \mathbb{Z}/2\mathbb{Z} \times V \to V$ such that $(V, +, \odot)$ is a $\mathbb{Z}/2\mathbb{Z}$-vector space.

   (b) Deduce that $(\mathcal{P}(X), \triangle)$ is a $\mathbb{Z}/2\mathbb{Z}$-vector space.

5. Recall that a field K has *characteristic* $p \geq 1$ iff p is minimal such that $\underbrace{1 + 1 + \cdots + 1}_{p \text{ times}} = 0$, or has *characteristic zero* if no such p exists.

   (a) Suppose K has characteristic zero. Prove that there is an injective map $\iota : \mathbb{Q} \to K$ such that $\iota(0) = 0_K$, $\iota(1) = 1_K$, $\iota(x + y) = \iota(x) + \iota(y)$, $\iota(xy) = \iota(x)\iota(y)$ for all $x, y \in \mathbb{Q}$.

   Hence all fields of char. zero can be viewed as extensions of $\mathbb{Q}$.

   (b) Suppose K has characteristic $p > 0$. Prove that p is prime.

6. Let V a K-vector space and $A \subseteq V$. Prove that $\text{span}(A) = A$ if and only if, for every $\alpha \in K$, $v, w \in A$, we have $\alpha v + w \in A$.

7. Prove that the span is a *closure operator*, i.e.

   (*i*) $A \subseteq \text{span}(A)$ for all $A \subseteq V$,

   (*ii*) $A \subseteq B \Rightarrow \text{span}(A) \subseteq \text{span}(B)$,

   (*iii*) $\text{span}(\text{span}(A)) = \text{span}(A)$.

8. Prove that $\text{span}(A)$ is the smallest set containing A and closed under $+, \cdot$.

9. Let ℓ^2 denote the space of square-summable real sequences $(x_n)_n$, i.e. sequences $(x_n)_n$ in $\mathbb{R}$ such that $\sum_{n=1}^{\infty} |x_n|^2 < \infty$. You may take for granted (for now) that ℓ^2 is a vector space over $\mathbb{R}$ equipped with the natural operations. Let $e^{(j)}$ be the sequence whose jth term is 1 and all other terms are zero.

   Is $\text{span}\{e^{(j)} : j \in \mathbb{N}\} = \ell^2$? Prove or disprove it.

10. Prove that the space of polynomials $\mathbb{R}[x]$ has no finite generating set.

11. Let $\mathbb{R}[[x]]$ denote the space of *formal power series*, i.e. expressions of the form $\sum_{n=0}^{\infty} a_n x^n$ for coefficients $a_i \in \mathbb{R}$. Assuming this is a vector space, show that $\{x^n : n \in \mathbb{N}\}$ does *not* span $\mathbb{R}[[x]]$.

3 Independence, linear maps: duality

1. Let V a K-VS. Prove that $I \subseteq V$ is independent if and only if every finite subset of I is independent.

2. (a) Find all vector spaces Z such that for any vector space W there is exactly one linear map $Z \to W$.
   (b) Find all vector spaces Z such that for any vector space V there is exactly one linear map $V \to Z$.

3. Recall that for a K-vector space V, its *dual space* is exactly $\hom(V, K)$. Let V be a vector space, $I = \{v_n : n \in \mathbb{N}\}$ a set of vectors in V. If there is a set of linear maps $\{f_n : n \in \mathbb{N}\} \subseteq V^*$ such that $f_m(v_n) = 1$ if and only if $m = n$ and 0 otherwise, prove that I is independent.

   Such a set of $\{f_n : n \in \mathbb{N}\}$ is called a *dual system* for $\{v_n : n \in \mathbb{N}\}$.

4. Consider the vector space $C^\infty(\mathbb{R}; \mathbb{R}) = \{f : \mathbb{R} \to \mathbb{R} \text{ infinitely differentiable}\}$. Let P denote the positive primes and define
$$I = \left\{ \sin\left(\frac{2\pi}{p} x\right) : p \in P \right\}.$$
   Use (2) to prove that I is independent.

5. Let V, W be K-VSes, $T_1, T_2 : V \to W$ linear. Let $S \subseteq V$ spanning. If $T_1\big|_S = T_2\big|_S$, prove that $T_1 = T_2$.

6. Let V be a K-vector space, $B \subseteq V$ a basis, $\iota : B \to V$ the inclusion map. Prove that for any vector space W and $f : B \to W$ there is a unique $T : V \to W$ linear such that $f = T \circ \iota$.

7. If V, W are K-vector spaces, $T : V \to W$ linear, prove that $\ker(T)$ and $\operatorname{im}(T)$ are subspaces of V and W respectively.

8. Let V, W be vector spaces over some field K and $T : V \to W$ linear.
   (a) If T is injective, prove that $T[I]$ is independent for every $I \subseteq V$ independent.
   (b) If T is surjective, prove that $T[S]$ is spanning for every $S \subseteq V$ spanning.

9. Let V be a K-VS and $I = \{v_1, \ldots, v_n\} \subseteq V$ independent. Prove that there is a dual system $\{f_1, \ldots, f_n\}$ for I, but that in general this dual system might not be unique.

10. If V is finite-dimensional, prove that every basis B admits a unique dual system B^*. Furthermore, show that this dual system is itself a basis of V^*.

11. Recall that for a vector space V, we define the double-dual $V^{**} := (V^*)^*$.
    (a) For V a K-VS, $v \in V$, let $\mathrm{ev}_v : V^* \to K$ via $\mathrm{ev}_v(f) = f(v)$. Prove that $\mathrm{ev}_v \in V^{**}$.
    (b) Define $\varphi : V \to V^{**}$ via $\varphi(v) = \mathrm{ev}_v$. Prove that φ is an injective linear map and deduce that V is isomorphic to a subspace of V^{**}.
    (c) If V is finite-dimensional, show that φ is an isomorphism from V onto V^{**}.

12. Let V be a vector space, $S \subseteq V$ independent. Consider the set

$$P = \{A \subseteq V \mid A \text{ independent}, A \supseteq S\}.$$

(a) Let $\mathcal{C} \subseteq P$ be such that, for any $I, J \in \mathcal{C}$, we have $I \subseteq J$ or $J \subseteq I$ (inclusive).
Prove that $\bigcup_{C \in \mathcal{C}} C$ is independent.

(b) By a result of set theory known as Zorn's lemma, from part (a) we are able to deduce that there is $\hat{I} \in P$ *maximal under inclusion*. That is, if $J \in P$ with $\hat{I} \subseteq J$, then $\hat{I} = J$.
Prove that $\hat{I}$ is a basis of V containing S.

(c) Let $f : \mathbb{R} \to \mathbb{R}$ satisfy *Cauchy's functional equation*:

$$f(x+y) = f(x) + f(y), \quad f(1) = 1. \tag{$*$}$$

Prove that f is $\mathbb{Q}$-linear, i.e. f is linear in the vector space $\mathbb{R}$ over $\mathbb{Q}$.

(d) Prove that solutions to $(*)$ exist which are not the identity.

13. Let V be a finite-dimensional vector space over $\mathbb{R}$ and suppose there is $J : V \to V$ linear such that $J^2(v) = -v$ for all $v \in V$.

(a) Let $\mathcal{L}$ be an ordered basis of V and $A = [J]_{\mathcal{L}}^{\mathcal{L}}$. Prove that $A^4 = I_n$.

(b) Define a multiplication $\odot : \mathbb{C} \times V \to V$ via

$$(a + bi) \odot v = av + bJ(v).$$

Prove that $V_J := (V, +, \odot)$ is a $\mathbb{C}$-vector space. Use this fact to prove that $\dim V$ is even.

(c) Prove that

$$\hom(V_J, V_J) = \{T \in \hom(V, V) \mid T \circ J = J \circ T\}.$$

(d) Let $W \subseteq V$. Prove that W is a subspace of V_J if and only if W is a subspace of V and $J[W] = W$.

14. Let V, W, U be vector spaces over some field K. We say a function $f : V \times W \to U$ is *billinear* if and only if f is linear in each argument separately. That is:

(i) for every $v \in V$ the map $w \mapsto f(v, w)$ is a linear map $W \to U$,

(ii) for every $w \in W$ the map $v \mapsto f(v, w)$ is a linear map $V \to U$.

Let $\hom_2(V \times W, U)$ be the space of all billinear maps $f : V \times W \to U$.

(a) Prove that $\hom_2(V \times W, U)$ is a subspace of $U^{V \times W}$.

(b) Show that neither $\hom_2(\mathbb{R}^2, \mathbb{R})$ nor $\hom(\mathbb{R}^2, \mathbb{R})$ is a subset of the other.

(c) Let $\varphi : V \times \hom(V, K) \to K$ via $\varphi(v, f) = f(v)$. Prove that φ is billinear.

(d) Suppose the dimension of V, W, U are all finite, equal to n, m, k respectively. Show that

$$\dim(\hom_2(V \times W, U)) = n \cdot m \cdot k.$$

Hint: It may help to consider the case where $U = K$ first.

15. *A genuinely absurd exercise.* Let Λ be an infinite indexing set, V an infinite-dimensional K-vector space with basis indexed by Λ. Prove that

$$\dim V^* = |K^\Lambda| = |V^*|,$$

and deduce that the algebraic dual of an infinite-dimensional vector space is in general too large to do anything useful with. (Google *Erdös-Kaplansky theorem.*)

4 Coordinates and matrices

1. Let $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map which acts as a rotation an angle θ anticlockwise about the origin. Compute the matrix of R_θ with respect to the standard basis, and deduce angle addition formulae for the cosine and sine.

2. Compute the matrix of the projection $P : \mathbb{R}^3 \to \mathbb{R}^2$ via $P(x, y, z) = (y, z)$, with respect to:

   (a) the standard bases of $\mathbb{R}^3$ and $\mathbb{R}^2$,

   (b) the bases $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (1, 7)\}$ of $\mathbb{R}^2$.

3. Let $V = C^\infty(\mathbb{R}; \mathbb{R})$, $r, k > 0$, $S_{k,r} = \operatorname{span}\{\sin(kx), \cos(kx), e^{-rx}\}$. Let $D : S_{k,r} \to V$ via $D(f) = \frac{df}{dx}$.

   (a) Verify that D can be viewed as a linear map from $S_{k,r}$ to itself. From here on, we restrict D to this space.

   (b) Show that $\{\sin(kx), \cos(kx), e^{-rx}\}$ forms a basis of $S_{k,r}$.

   (c) Determine the matrix of D with respect to this basis.

4. Prove that for every elementary row operation there is a unique $n \times n$ matrix E such that, for any $n \times m$ matrix A, EA is the result of applying the corresponding row operation to A.

5. (a) Invert $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K^{2\times 2}$ by Gaussian elimination, where $a, b, c, d \in K$ and $ad - bc \neq 0$.

   (b) If a 2×2 matrix has integer entries and $ad - bc = \pm 1$, prove that its inverse has integer entries.

6. How many invertible 3×3 matrices are there in $(\mathbb{Z}/7\mathbb{Z})^{3\times 3}$?

7. Let $x_0, \ldots, x_n \in \mathbb{R}$ and $\varphi : P_n(\mathbb{R}) \to \mathbb{R}^{n+1}$ via $\varphi(p) = (p(x_0), p(x_1), \ldots, p(x_n))$. Prove that φ is bijective if and only if all x_i are distinct.

8. Use Gaussian elimination over the field $\mathbb{Z}/11\mathbb{Z}$ to solve the system of congruences:

$$\begin{cases} 3x + y \equiv 1 \pmod{11} \\ x + 2y \equiv 6 \pmod{11} \end{cases}$$

9. Let V be a vector space. We say that V satisfies the *ascending chain condition*, or ACC, if and only if every increasing sequence of subspaces of V stabilizes. That is, if $V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots$ is an increasing sequence of subspaces, then there is $N \in \mathbb{N}$ such that, for all $i, j \geq N$, $V_i = V_j$.

   (a) Prove that a vector space V satisfies the ACC if and only if it is finite-dimensional.

   (b) Deduce that if A is an $n \times n$ matrix then the sequence of subspaces $\{\ker(A^t)\}_{t\in\mathbb{N}}$ stabilizes.

10. (a) Find the rank and nullity of the (real) matrix

$$\begin{pmatrix} 4 & 5 & 6 \\ 8 & 10 & 12 \\ 12 & 15 & 18 \end{pmatrix}.$$

   (b) Let $u, v \in \mathbb{R}^n$ both nonzero and define $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ via

$$a_{ij} = u_i v_j.$$

   Prove that $\operatorname{rank}(A) = 1$ and deduce the nullity of A.

   (c) If $A \in \mathbb{R}^{n\times n}$ such that $\operatorname{rank}(A) = 1$, are there $u, v \in \mathbb{R}^n$ nonzero such that $A = uv^T$? Prove it, or find a counterexample.

11. Let $S \in K^{n\times n}$ be the $n \times n$ matrix whose i, j-entry is given by

$$s_{ij} = \begin{cases} 1 & i+1=j \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the smallest $m \in \mathbb{N}$ such that $S^m = 0$, with proof. Denote this value by N.

(b) If $v \in K^n$ with $v_n \neq 0$, show that $\{S^t v : 0 \leq t < N\}$ is an independent set. Is it a basis?

(c) Call a matrix $A = (a_{ij}) \in K^{n\times n}$ *strictly upper triangular* if and only if $a_{ij} = 0$ for all $j \leq i$. Find the smallest $m \in \mathbb{N}$ such that $A^m = 0$ for all strictly upper triangular matrices A, with proof.

12. Recall that $\mathbb{R}^{n\times n}$ forms a vector space over $\mathbb{R}$. The *trace* $\operatorname{tr} : \mathbb{R}^{n\times n} \to \mathbb{R}$ is defined via

$$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}.$$

(a) Prove that tr is linear and $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for all $A, B \in \mathbb{R}^{n\times n}$.

(b) Let

$$\mathcal{C} = \{C \in \mathbb{R}^{n\times n} \mid \text{there are } A, B \in \mathbb{R}^{n\times n} \text{ with } C = AB - BA\}.$$

Prove that $\operatorname{span} \mathcal{C} = \{M \in \mathbb{R}^{n\times n} \mid \operatorname{tr}(M) = 0\}$.

(c) Let $f : \mathbb{R}^{n\times n} \to \mathbb{R}$ be linear such that $f(AB) = f(BA)$ for all $A, B \in \mathbb{R}^{n\times n}$. Use part (b) to prove that there is $\lambda \in \mathbb{R}$ such that $f(A) = \lambda \cdot \operatorname{tr}(A)$ for all $A \in \mathbb{R}^{n\times n}$.

5 Determinants and diagonalization

In this section, if A is an $n \times n$ complex matrix, we define

$$\sigma(A) = \{\lambda \in \mathbb{C} \mid \operatorname{null}(A - \lambda I_n) > 0\}.$$

1. Compute the determinant and trace of the rotation R_θ of an angle θ counterclockwise about some nonzero $x \in \mathbb{R}^3$.

2. Let $x_0, \ldots, x_n \in K$ and consider the $(n+1) \times (n+1)$ matrix $V(x_0, \ldots, x_n)$, whose (i,j)-entry is given by
$$v_{ij} = x_{i-1}^{j-1}.$$
Compute $\det V(x_0, \ldots, x_n)$ by expressing V as the matrix of some nice linear map.

3. Let F_n be the nth Fibonacci number. Use linear-algebraic methods to prove that, for every $n \geq 1$,
$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n.$$
(This is known as *Cassini's identity.*)

4. Let $a_0, a_1 \in \mathbb{R}$ given, $\lambda, r \in \mathbb{R}$. For $n \geq 2$, define $a_n = \lambda a_{n-1} + r a_{n-2}$. Exhibit a Cassini-type identity for this sequence, i.e. find a closed form for
$$a_{n+1}a_{n-1} - a_n^2,$$
for $n \geq 1$.

5. Let $A \in \mathbb{C}^{n \times n}$.
   (a) Prove that
   $$\sigma(A) = \{\lambda \in \mathbb{C} \mid \lambda \text{ is an eigenvalue of } A\},$$
   and deduce that $\sigma(A) \neq \varnothing$.
   (b) Show that it is possible for $\sigma(A) \cap \mathbb{R} = \varnothing$.

6. Prove that the following are equivalent for a square matrix $A \in \mathbb{C}^{n \times n}$:
   (*i*) $A^2 = A$,
   (*ii*) A is diagonalizable and $\sigma(A) \subseteq \{0, 1\}$.

7. Prove that, for $A \in \mathbb{C}^{n \times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ (with multiplicity), we have
$$\det(A) = \prod_{i=1}^{n} \lambda_i.$$

8. Let $f : \mathbb{C} \to \mathbb{C}$ be defined everywhere by a power series:
$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$
   (a) If $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ is diagonal, prove that $f(D) = \operatorname{diag}(f(\lambda_1), \ldots, f(\lambda_n))$.
   (b) If $A \in \mathbb{C}^{n \times n}$ is diagonalizable, compute $f(A)$.
   (c) Compute
   $$\exp \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

6 Inner product and normed spaces

1. Let $V = C(S^1; \mathbb{C})$ denote the continuous functions $f : S^1 \to \mathbb{C}$, where S^1 is the unit circle in $\mathbb{C}$. Easily, this is a vector space over $\mathbb{C}$.

   (a) Prove that
   $$\langle f, g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\overline{g(e^{i\theta})}\, d\theta$$
   is an inner product on V.

   (b) For $n \in \mathbb{Z}$, let $f_n : S^1 \to \mathbb{C}$ via $f_n(z) = z^n$. Prove that $\{f_n : n \in \mathbb{N}\}$ is an orthonormal set.

2. Let A a finite set, $\langle \cdot, \cdot\rangle_{ev} : \mathbb{C}^A \to \mathbb{C}^A$ via
   $$\langle f, g\rangle_{ev} = \sum_{a\in A} f(a)\overline{g(a)}.$$

   (a) Prove that this is an inner product.

   (b) For $W \subseteq A$, let χ_W be its characteristic function. The expression $\langle \chi_W, \chi_U\rangle_{ev}$ has combinatorial meaning for $W, U \subseteq A$. What is it?

3. Let $G = (V, E)$ be a graph. Associated to this graph is an inner product space $(\mathbb{R}^V, \langle \cdot, \cdot\rangle_{ev})$) and an *adjacency operator* $A : \mathbb{R}^V \to \mathbb{R}^V$ given by
   $$(Af)(v) = \sum_{\{v,w\}\in E} f(w).$$

   (a) Justify that A is linear.

   (b) Describe the matrix of A w.r.t. the usual basis of δ-functions.

   (c) Let $U, W \subseteq V$. The expression $\langle A\chi_U, \chi_W\rangle$ has meaning; what is it?

   (d) Suppose every vertex of G has the same number of neighbors d. What can you say about the eigenvalues of A?

4. Let $\{e_1, \ldots, e_n\}$ be an orthonormal basis of an inner product space V. Show that
   $$v = \sum_{i=1}^{n} \langle v, e_i\rangle e_i$$
   and deduce that
   $$\|v\| = \sum_{i=1}^{n} |\langle v, e_i\rangle|^2.$$

5. Given a subspace A of a *finite-dimensional* real inner product space V, define
   $$A^\perp = \{v \in V \mid \langle a, v\rangle = 0 \text{ for all } a \in A\}.$$

   (a) Prove that $A^\perp$ is a subspace of V.

   (b) Show that $V = A + A^\perp$ and $A \cap A^\perp = \{0\}$.

   (c) Show that $(A^\perp)^\perp = A$.

6. Let V be a finite-dimensional inner product space over $\mathbb{R}$. Prove that for every $f \in V^*$ there is a unique $w \in V$ such that $f(v) = \langle w, v\rangle$ for all $v \in V$. (This is a *Riesz-Representation* theorem.)

Apparently y'all didn't define normed spaces in lecture, so I'll record its definition here.

Definition. *Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. A* ***norm*** *on V is a map $\|\cdot\| : V \to [0, \infty)$ such that:*

(i) $\|v\| = 0$ *iff* $v = 0$,

(ii) $\|\lambda v\| = |\lambda| \, \|v\|$ *for all* $v \in V$, $\lambda \in \mathbb{F}$,

(iii) $\|v + w\| \leq \|v\| + \|w\|$ *for all* $v, w \in V$.

The pair $(V, \|\cdot\|)$ is a ***normed space****.*

You've already proven that, for example, every inner product space is a normed space with norm

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

But not every norm comes from an inner product[1], so you can't assume a general norm takes this form.

7. Let $(V, \|\cdot\|_V), (W, \|\cdot\|_W)$ be normed spaces. We say a linear map $T : V \to W$ is *bounded* provided that

$$\|T\|_{op} = \sup_{\|v\|_V = 1} \|Tv\|_W < \infty.$$

   Prove that $\{T \in V^* \mid \|T\|_{op} < \infty\}$ forms a subspace of V^*. This is known as the *continuous dual* and is often denoted by V'.

8. Let V be a *finite-dimensional* real inner product space. You've already shown that if we define $\|v\| = \sqrt{\langle v, v \rangle}$, then $(V, \|\cdot\|)$ is a normed space. It turns out that $(V', \|\cdot\|_{op})$ is also a normed space; you need not prove this. Given $v \in V$, let $\varphi(v) \in V^*$ via

$$\varphi(v)(w) = \langle v, w \rangle.$$

   (a) Prove that $\varphi : V \to V'$, i.e. $\varphi(v)$ is bounded for every v. Deduce that $V' = V^*$.

   (b) By problem 6, φ is an isomorphism. Prove that φ is furthermore an isometry, i.e.

$$\|\varphi(v)\|_{op} = \|v\|$$

   for all $v \in V$. Is φ bounded as a map $V \to V'$?

9. If V is an infinite-dimensional normed space, prove $V^* \neq V'$.

[1] Google "parallelogram law."

7 Adjoints and spectral theory

1. Let V an inner product space, $T : V \to V$ normal. Prove that $\ker(T) = \ker(T^*)$ and that T, T^* have the same eigenvalues.

2. Let $A, B \in \mathbb{C}^{n\times n}$. Prove that
$$\ker(A^*A + B^*B) = \ker(A) \cap \ker(B).$$

3. Let $V = \{f \in C^\infty([0,1];\mathbb{R}) \mid f^{(n)}(0) = f^{(n)}(1) = 0 \text{ for all } n \in \mathbb{N}\}$ be the space of smooth functions $f : [0,1] \to \mathbb{R}$ such that f and all derivatives vanish at the endpoints. Determine the adjoints of the following maps with respect to the integral inner product, or prove that they don't exist.
   (a) $T_\lambda : V \to V$ via $(T_\lambda f)(x) = f(\lambda x)$, $\lambda \in [0,1)$,
   (b) $D : V \to V$, $(Df)(x) = f'(x)$,
   (c) $p(D) : V \to V$ where D is as in (b) and p is any non-constant polynomial.

4. Let $\langle \cdot, \cdot \rangle : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ via
$$\langle A, B\rangle = \operatorname{tr}(B^*A).$$
   Prove that $\langle \cdot, \cdot \rangle$ is an inner product.

5. Consider the map $F : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ given by $F(A) = A^T$.
   (a) It follows from (4) that the map
$$\langle A, B\rangle = \operatorname{tr}(B^TA)$$
   is an inner product on $\mathbb{R}^n$. Prove that $F^* = F$ with respect to this inner product.
   (b) Prove the eigenvalues of F are exactly 1 and -1.
   (c) Give natural descriptions of the eigenspaces E_1 and E_{-1}, and perform a computation to verify that the eigenspaces are orthogonal.
   (d) Find an orthonormal basis of $\mathbb{R}^{n\times n}$ consisting of eigenvectors for F.

6. Let T be an operator on a complex inner product space. Prove that T is diagonalizable iff $[T]_{\mathcal{M}}^{\mathcal{M}}$ is diagonalizable for some $\mathcal{M}$ a basis of V iff $[T]_{\mathcal{M}}^{\mathcal{M}}$ is diagonalizable for every $\mathcal{M}$ a basis of V.

7. Let V be an infinite-dimensional vector space over $\mathbb{R}$. By problem 3 on the previous blue-book practice exam, V still admits a basis in the usual sense; let this basis be $\mathcal{B} = \{b_\alpha : \alpha \in \Lambda\}$.
   (a) Let $B : \mathcal{B} \times \mathcal{B} \to \mathbb{R}$ be any function. Prove that there is a unique billinear map $\tilde{B} : V \times V \to \mathbb{R}$ such that $\tilde{B}\big|_{\mathcal{B}\times\mathcal{B}} = B$.
   (b) Show that there is an inner product $\langle \cdot, \cdot \rangle_{\mathcal{B}}$ on V such that $\mathcal{B}$ is orthonormal in $(V, \langle \cdot, \cdot \rangle_{\mathcal{B}})$.

8. Say that a symmetric matrix $A \in \mathbb{R}^{n\times n}$ is *positive semidefinite* (PSD) iff $\langle Ax, x\rangle \geq 0$ for all $x \in \mathbb{R}^n$. Let $A \in \mathbb{R}^{n\times n}$ symmetric.
   (a) Prove that A is PSD iff all eigenvalues of A are nonnegative.
   (b) Suppose A is positive semidefinite. Prove that there is a unique positive semidefinite $B \in \mathbb{R}^{n\times n}$ such that $B^2 = A$.

8 More spectral theory and singular value decomposition

1. Consider
$$A = \begin{pmatrix} 5 & 1 \\ -1 & -5 \\ -1 & -5 \\ 5 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 \\ 3 & 4 & 5 & 6 & 7 \end{pmatrix}.$$
   (a) Let $A = U\Sigma V^*, B = LDR^*$ be SVDs for A, B respectively. What are the last two rows of Σ? Last two columns of D?
   (b) Compute an SVD for A.

2. Let V, W complex inner product spaces, $T : V \to W$. Show that TT^* and T^*T have the same nonzero eigenvalues with multiplicity.

3. Let $A \in \mathbb{R}^{n\times n}$. Prove that there are U orthogonal and P positive semidefinite such that $A = UP$. Show that P is unique, and if A is invertible then U is also unique. (This is known as *polar decomposition.*)

4. Prove that for every $n \times n$ positive semidefinite matrix $A \in \mathbb{R}^{n\times n}$ there are n vectors v_i such that
$$A = \sum_{i=1}^{n} v_i v_i^T.$$
   If A is not positive semidefinite, how can you generalize this result?

5. (*Gershgorin's circle theorem.*) Let $A = (a_{ij}) \in \mathbb{C}^{n\times n}$. For $i \in [n]$, define $R_i = \sum_{j\neq i} |a_{ij}|$. Let $D(a_{ii}, R_i) \subseteq \mathbb{C}$ be a closed disk of radius R_i centered at a_{ii}.
   (a) Prove that every eigenvalue of A lies in at least one $D(a_{ii}, R_i)$.
   (b) Suppose A is such that
$$|a_{ii}| \geq \sum_{j\neq i} |a_{ij}|$$
   for all i; such an A is called *diagonally dominant,* and is called *strictly* diagonally dominant iff this inequality is strict. If A is strictly diagonally dominant, prove that A is invertible.
   (c) Let A be a self-adjoint complex matrix with real nonnegative diagonal entries. If A is diagonally dominant, prove that $\langle Az, z\rangle$ is real and nonnegative for every $z \in \mathbb{C}^n$.

6. *On matrix decompositions used in numerical analysis.*
   (a) Let $A \in \mathbb{C}^{n\times n}$. Prove that we can factor $A = QR$ where Q is unitary and R is upper triangular. This is the QR decomposition, unsurprisingly.

   Hint: Tenz-Fpuzvqg.
   (b) Let $A \in \mathbb{C}^{n\times n}$. Prove that we can factor $A = LL^*$ where L is lower triangular.

   This has a name: *Cholesky decomposition.*

7. (*Spectral mapping theorem.*) Let $\sigma(A)$ denote the complex eigenvalues of $A \in \mathbb{C}^{n\times n}$, $p \in \mathbb{C}[x]$. Prove that
$$\sigma(p(A)) = p[\sigma(A)].$$

8. Corollaries of spectral mapping.
   (a) Deduce that a nilpotent matrix has 0 as its only eigenvalue.
   (b) Prove that $\sigma(\exp(A)) = \exp[\sigma(A)]$.

9. Some localized definitions.

- An *abstract projection matrix* is a complex matrix P such that $P^2 = P$.
- We say $A \in \mathbb{C}^{n\times n}$ has the PF-property iff there is an eigenvalue $r \in \mathbb{R}$ of algebraic multiplicity 1 such that $|\lambda| < |r|$ for all other eigenvalues λ of A. Call r the PF-eigenvalue of A.

(a) If P is an abstract projection matrix, prove that $\operatorname{tr}(P) = \operatorname{rank}(P)$.

Hint: Erpnyy sebz erpvgngvba gung na nofgenpg cebwrpgvba vf qvntbanyvmnoyr jvgu nyy rvtrainyhrf mreb be bar.

(b) Let $A \in \mathbb{C}^{n\times n}$ have the PF-property with r the PF-eigenvalue, v an eigenvector of A for r, w an eigenvector of A^T for r all normalized so that $w^T v = 1$. The rest of this problem will be working under this framework.

Prove that $P := \lim_{k\to\infty} A^k/r^k$ exists entrywise.

(c) Show that P is an abstract projection and $PA = AP$.

(d) Show that $\operatorname{rank}(P) = \operatorname{rank}(P^T) = 1$, $\operatorname{im} P$ is spanned by v, and $\operatorname{im} P^T$ is spanned by w.

(e) Conclude that $P = vw^T$.

10. We say a matrix $A \in \mathbb{R}^{n\times n}$ is *stochastic* iff all of the following hold:

- each of its columns sum to 1,
- each of its entries are between 0 and 1 inclusive,
- every row has a nonzero entry,
- some column has more than one nonzero entry.

In addition, we say A is *regular stochastic* iff A is stochastic and A^n has all positive entries for some n.

[This one's almost ridiculous.] Prove that regular stochastic matrices enjoy the PF-property with $r = 1$. (This result, bluntly, makes Markov chains worth using.)

11. *An abstract alternate version of the spectral theorem.* Let T be a self-adjoint linear operator on a finite-dimensional complex inner product space, with eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$. Show that

$$T = \sum_{i=1}^{n} \lambda_i P_i,$$

where $P_i^2 = P_i = P_i^*$ and $\sum_{i=1}^n P_i = 1$. Compute this decomposition when $T : \mathbb{C}^3 \to \mathbb{C}^3$ is given by

$$T(v) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -i & -1 \\ 1 & -1 & 0 \end{pmatrix} v.$$

12. *A handwavy, but cool, thing.* On the space $\mathcal{S}(\mathbb{R})$ of smooth $f : \mathbb{R} \to \mathbb{R}$ which decay rapidly at infinity, the *Fourier transform*

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x}\, dx$$

makes sense and is very well-behaved; it even has an inverse $\mathcal{F}^{-1}$.

Computing with reckless abandon (i.e. don't worry about analytic justifications here), show that the Fourier transform *diagonalizes differentiation.* That is, show that there is $\lambda \in \mathbb{C}$ such that

$$D = \mathcal{F}^{-1} \circ T_\lambda \circ \mathcal{F},$$

where $D : \mathcal{S}(\mathbb{R}) \to \mathcal{S}(\mathbb{R})$ is the derivative[2] and $T_\lambda(v) = \lambda v$.

[2] You may assume I'm not lying to you in this specification, i.e. the derivative

9 Some fun with quotients

Before Thanksgiving, I held a special topic recitation on quotients. This content cannot be directly tested, but they are useful exercises to really hammer in the basics.

These first few exercises demonstrate that if V is an arbitrary vector space over an arbitrary field, then every $W \subseteq V$ is naturally the kernel of a linear map (i.e. without referring to a basis), and we deduce a few isomorphism theorems as a result of this. In this section, V, W could be infinite-dimensional, or over any field.

1. Let V a vector space, $W \subseteq V$ a subspace. Define an equivalence relation $\sim$ on V via

$$v \sim v' \iff v - v' \in W.$$

   Denote the equivalence class of v under $\sim$ by $[v]$, and the set of classes by $V/\!\sim$. We attempt to define $+ : V/\!\sim \times V/\!\sim \to V/\!\sim$ and $\cdot : K \times V/\!\sim \to V/\!\sim$ via

$$[v] + [v'] := [v + v']$$

   and

$$\alpha[v] := [\alpha v].$$

   Prove that these operations are well-defined. These give $V/\!\sim$ the structure of a vector space; viewed in this way, we write this structure as V/W, colloquially "V mod W."

2. There is a natural map $q_W : V \to V/W$ via $q(v) = [v]$. Prove that q_W is linear and $\ker(q_W) = W$.

3. Let $\varphi : V \to W$ linear, $S \subseteq \ker(\varphi)$ a subspace. Prove that there is a unique linear map $\tau : V/S \to W$ such that $\tau \circ q_S = \varphi$, and for this τ we have $\operatorname{im} \tau = \operatorname{im} \varphi$, $\ker(\tau) = \ker(\varphi)/S$.

4. By taking S appropriately in (3), show that if $\varphi : V \to W$ is linear then $V/\ker(\varphi) \cong W$. This is known as (an instance of) the *First Isomorphism Theorem*. Deduce that if $\varphi : V \to W$ is linear, surjective then W is isomorphic to a quotient of V.

5. Other isomorphism theorems. We didn't prove these, but they're decent if challenging exercises.

   (a) Let V a vector space, $S, T \subseteq V$ subspaces.

      (i) Prove that $S \cap T$ is a subspace of S.

      (ii) Prove that T is a subspace of $S + T = \{s + t : s \in S, t + T\}$.

      (iii) Prove that

$$(S + T)/T \cong S/(S \cap T).$$

      [Spam problem 4.]

   (b) Let V a vector space, $T \subseteq S \subseteq V$ subspaces.

      (i) Prove that S/T is a subspace of V/T.

      (ii) Prove that

$$(V/T)/(S/T) \cong V/S.$$

We now ask: what happens if we put the word "continuous" in front of everything? Obviously, nothing involving continuity is testable.

Definition. *We say a subset S of a normed space $(V, \|\cdot\|)$ is* ***closed*** *iff, whenever $\{v_n\}_n$ is a sequence in S with $v_n \to v$ in V, we have $v \in S$. (For a definition of $v_n \to v$, see the red-book midterm 2 practice.)*[3]

6. Let $(V, \|\cdot\|_V), (W, \|\cdot\|_W)$ be normed spaces and $\varphi : V \to W$ a bounded linear map. Prove (using the definitions of boundedness and limit we've discussed) that $\ker \varphi$ is closed.

7. Recall the not-actually-a-basis $e^{(i)}$ of $\ell^2(\mathbb{C})$ again. Prove that $\operatorname{span}\{e^{(i)} : i \in \mathbb{N}\}$ is *not* closed.

So, in particular, we have a nontrivial condition on kernels of bounded linear maps: they're always *closed subspaces*. Does the converse hold, i.e. is every closed subspace the kernel of a bounded linear map? Magically, the answer is *yes*!

8. Let $(V, \|\cdot\|)$ be a normed space and W a *closed* subspace of V. Consider the map $\|\cdot\|_q : V/W \to [0, \infty)$ given by
$$\|[v]\|_q = \inf\{\|v + w\| : w \in W\}.$$
Prove that $\|\cdot\|_q$ is well-defined. (Recall that the infimum of a subset of $\mathbb{R}$ is the greatest lower bound of the set, and that any subset of $\mathbb{R}$ bounded below has an infimum.)

9. Prove that $\|\cdot\|_q$ is a norm. [You *will* need to use the fact that W is closed to show definiteness.]

10. Let $q_W : V \to V/W$ via $q_W(v) = [v]$ be the natural quotient (linear) map from earlier. Prove that q_W is bounded as a map $(V, \|\cdot\|) \to (V/W, \|\cdot\|_q)$.

Hence $W = \ker(q_W)$ is the kernel of a bounded linear map, as desired.

[3] For the astute, you will note that this is not the most general definition of closed. But this is still a linear algebra course, and this is the definition which is most useful for now.

10 Some applications of JCF, and other problems

1. Let J be a Jordan block. Show that the difference between the algebraic and geometric multiplicities of the eigenvalues of J are maximized.

2. Let $A \in \mathbb{C}^{n\times n}$, and define
$$\exp(A) = \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$
Also, denote
$$\|A\|_\infty := \max_{(i,j)\in[n]^2} |a_{ij}|.$$
   (a) Show that $\|\cdot\|_\infty$ is a norm on $\mathbb{C}^{n\times n}$.
   (b) Prove that the series defining $\exp(A)$ converges entrywise.
   (c) If $AB = BA$, prove that $\exp(A+B) = \exp(A)\exp(B)$. Find small examples to show that this can fail when $AB \neq BA$.
   (d) Compute $\exp(J)$ when J is a Jordan block.
   (e) Describe how this can be used to compute $\exp(A)$ for any $A \in \mathbb{C}^{n\times n}$.

3. Let X be the space of 2×2 complex matrices of (real) positive trace and (real) positive determinant. For every $A, B \in X$, show that there is a map $\gamma : [0,1] \to X$ continuous with $\gamma(0) = A, \gamma(1) = B$.[4]

4. Show that every invertible operator on a finite-dimensional complex vector space has a complex square root, but $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ has no real square root.

[4] That is, find a γ which is sufficiently nice in t to be "obviously continuous".

11 Extra questions

1. Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. A *seminorm* on a general $\mathbb{F}$-vector space V is a map $p : V \to [0, \infty)$ such that:

   (*i*) $p(\lambda v) = |\lambda| p(v)$ for all $v \in V$, $\lambda \in \mathbb{F}$

   (*ii*) $p(v + w) \leq p(v) + p(w)$ for all $v, w \in V$.

   The *kernel* $\ker p = \{w \in V \mid p(w) = 0\}$. It's clear that every norm is a seminorm, and if p is a seminorm then $\ker p = \{0\}$ iff p is a norm.

   (a) Prove that $\ker p$ is a subspace of V.

   (b) If $p, q : V \to [0, \infty)$ are seminorms, prove that $\max\{p, q\}$ is a seminorm.

   (c) Suppose V is a finite-dimensional complex inner product space, $W \subseteq V$ a subspace. Consider the map $p_W : V \to [0, \infty)$ via
   $$p_W(v) = \| \operatorname{proj}_W^{\perp}(v) \|,$$
   where $\| \cdot \|$ is the norm induced by the IP. Prove that p_W is a seminorm and $\ker p_W = W^{\perp}$. Conclude that every subspace W of V is the kernel of a seminorm.

   (d) Suppose V is general again. Let $p : V \to [0, \infty)$ a seminorm and $q_p : V \to V/\ker p$ the natural quotient map. Show that there is a unique norm $\tilde{p}$ on $V/\ker p$ such that $\tilde{p} = p \circ q_p$.

   (e) Still in the general setting. Is every subspace of V the kernel of a seminorm?

2. Let $(X, \| \cdot \|)$ be a Banach space and $\{x_n\}_n$ a sequence in X. If $\sum_{n=1}^{\infty} \|x_n\|$ converges in $\mathbb{R}$, prove that $\sum_{n=1}^{\infty} x_n$ converges in X. [5]

3. We investigate series of operators a little more closely. Let $(X, \| \cdot \|)$ be a Banach space and consider $\mathcal{B}(X)$ as a Banach space with the operator norm.

   (a) Prove that $\exp(T) := \sum_{n=0}^{\infty} \frac{T^n}{n!}$ converges in $\mathcal{B}(X)$ for every $T \in \mathcal{B}(X)$.

   (b) If $T, S \in \mathcal{B}(X)$ with $TS = ST$, prove that $\exp(T + S) = \exp(T)\exp(S)$.

   (c) Let $f : \mathbb{R} \to \mathbb{R}$ be defined via a power series which converges everywhere:
   $$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$
   Prove that $f(T)$ makes sense as an element of $\mathcal{B}(X)$. If f is bounded above by M, show that $\|f(T)\|_{op} \leq M$ for every $T \in \mathcal{B}(X)$.

4. Let V be a $\mathbb{C}$-vector space, $T : V \to V$ linear.

   (a) If V is finite-dimenisional, prove that there is $p \in \mathbb{C}[x]$ nonzero with $p(T) = 0$.

   (b) If V is infinite-dimensional but $\operatorname{im} T$ is finite-dimensional, prove that there is still some $p \in \mathbb{C}[x]$ nonzero with $p(T) = 0$.

[5] The converse is also true, i.e. absolute convergence implies convergence if *and only if* the underlying space is Banach. This is much harder to prove.